Fixed-Time Formation-Containment of Nonlinear Systems Using Intermittent Output and Connectivity

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***Abstract*—This article addresses the distributed fixed-time formation-containment problem of uncertain nonlinear systems considering intermittent output, connectivity, and actuation simultaneously. With the backstepping procedure, a unified fixed-time event-triggered formation-containment framework is established, achieving fast convergence while conserving resources. Different from existing fixed-time event-triggered results, the vexing issue of nondifferentiable virtual control laws caused by discontinuous outputs is resolved through a novel filter. Additionally, an asynchronous event-triggered fixed-time distributed estimator is constructed to reduce the communication cost among neighbors, eliminating the need for continuous monitoring and transmission of estimator states as seen in related results. Then, a fixed-time event-triggered control law without periodic actuation is proposed. The practical fixed-time stability of the closed-loop system is proven by adopting the Lyapunov theory. The feasibility of the explored control algorithm is illustrated by simulation and experiment involving unmanned aerial vehicles (UAVs) and unmanned ground vehicles (UGVs).**

***Index Terms*—Fixed-time event-triggered control, formation-containment control, nonlinear multiagent systems.**

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I. Introduction

The formation-containment control means that all leaders form a preset formation configuration guided by a virtual leader and the states of each follower are driven into a secure convex hull shaped by a group of leaders from the formation layer. Recently, the distributed formation-containment control of multiagent systems (MASs) has been a hot topic due to its great potential in engineering applications, such as the cooperative flight of multiple unmanned aerial vehicles and the swarm of hypersonic flight vehicles [1], [2], [3], [4], [5], [6]. Note that one of the most fundamental factors in performing the above control task lies in the robot platform, which is usually equipped with sensors, communication modules, microprocessors, and actuation modules. Generally speaking, the communication and energy resources deployed on the robot platform are limited [7], [8], [9], [10]. In addition, the settling time is a critical performance index for some urgent scenarios, such as cooperative rescue. Hence, it is desired to explore economical and efficient control algorithms to obtain superior performance while saving resources.

Considering the limited communication capability of the robot platform, where only a subset of leaders can obtain the virtual leader's information, a distributed observer-based formation-containment control strategy is developed for linear MASs [11]. This strategy avoids the global sharing of the virtual leader's information, but necessitates continuous communication among neighbors. Inspired by an event-triggered mechanism (ETM) [12], the output regulation issue of heterogeneous linear MASs is solved by designing an event-based adaptive distributed observer [13], where the communication frequency among neighbors is reduced. In [14], an event-driven distributed observer is constructed for cooperative control of nonlinear MASs to asymptotically estimate the state of the reference system. The convergence time of the above-mentioned results grow unboundedly as the initial observation errors increases. However, the priori information about the bound in its initial observation errors is unknown in most engineering scenarios. Moreover, for some time-critical tasks, such as multiple missile coordination, it is desired to realize fast estimation of the reference signal. Then, the fixed-time approach may severs as an effective solution in ensuring an upper bound on settling time regardless of initial cases [15], [16], [17], [18]. Note that fewer results explore distributed fixed-time event-triggered estimators

(DFTETEs) that strike a balance between estimation performance and resource conservation. Thus, how to develop a distributed fixed-time estimator with less network traffic burden for nonlinear MASs to guarantee practical fixed-time formation-containment remains a significant and open issue.

Based on the estimation of distributed observers, a local decentralized control algorithm is formulated for tracking the desired states. For further reducing the consumption of communication resources, the control scheme also is designed combining an event-triggered strategy. A self-triggered formation-containment controller is proposed for general linear MASs, relaxing the continuous monitoring requirement for measurement errors [19]. Inspired by [20] which incorporates an internal dynamic variate into the triggering threshold, a dynamic event-triggered mechanism is proposed for the leader-follower problem of nonlinear systems [21]. It is worth pointing out that for the above ETMs [19], [20], [21], continuous transmission of the control signal can be avoided in the controller-actuator channel. However, the measurement output still needs to be broadcast in the sensor-controller channel. Actually, energy resources of sensors to supply information distribution are also limited. More recently, the scheme of double event-triggered mechanisms is proposed for nonlinear systems with state triggering [22]. This approach guarantees asymptotic stabilization of the closed-loop system. However, it is important to acknowledge that discontinuous sampling actions or controller updates can negatively impact system performance. Hence, the prospect of a fixed-time event-triggered scheme arises, promising faster reaching time and stronger robustness. Meanwhile, the practical fixed-time stability theory is motivated by the fact that the ETMs inevitably affect the control performance, so that the tracking errors may fail to converge accurately to the origin within a fixed-time interval [23], [24]. Despite the proposals of fixed-time event-triggered control founded in [25] and [26], state triggering in the sensor-controller channel is not considered. Achieving intermittent output feedback within a practical fixed-time framework has not been well solved.

Moreover, the intermittent output signal within the sensor-controller channel brings some challenges for designing backstepping controller. To be specific, directly using discontinuous signals results in the nondifferentiability issue of virtual control laws. The backstepping control technique, as a nonlinear control paradigm, is widely applied in event-triggered control for nonlinear systems [27], [28], [29]. More recently, the output-triggered backstepping control scheme is proposed for MASs [30], [31], in which the closed-loop signals are uniformly ultimately bounded. Nevertheless, how to mitigate the impact of noncontinuous output on the derivatives of fixed-time virtual control laws is a challenging subject.

Inspired by the aforementioned discussion, we design a distributed fixed-time formation-containment control (DFTFCC) protocol via the backstepping method, where practical fixed-time stabilization is achieved despite involving discontinuous output, intermittent connectivity, and aperiodic execution. The main contributions are presented as follows.

1) Unlike the existing formation-containment control algorithms with periodic sampling setting [1], herein, under output triggering and intermittent communication, we first develop a DFTFCC law with aperiodic execution, reducing the communication traffic among neighbors, computation cost, and actuation frequency. Meanwhile, the effectiveness of the DFTFCC algorithm is verified on the UGVs platform.
2) In contrast to the fixed-time event-triggered control results based on the continuous measurement output [25], [26], herein, the virtual control laws are not differentiable under the impact of the output triggering. To address this issue within a fixed-time backstepping framework, a first-order low-pass filter and a new replacement function for the derivatives of the filtering output are introduced. These solutions effectively resolve the challenge encountered in the backstepping procedure and stability analysis. Different from the state-triggered backstepping control schemes [30], [31], the proposed approach ensures the practical fixed-time stability of the closed-loop system. Moreover, the setting time is explicitly given for the leader layer to shape a predefined geometrical configuration and each follower to reach the convex hull.
3) Compared with the distributed fixed-time observer-based control methods using continuous communication [18], the DFTETEs governed by an asynchronous ETM are respectively constructed for the leader layer and the follower layer. This allows for precise acquisition of desired position and velocity while eliminating the necessity for continuous monitoring and transmission of estimator states among neighbors.

II. Preliminaries and Problem Description

A. Graph Theory

The channel of message transmission among agents is characterized as a graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \Lambda)$ with node set $\mathcal{V} = \{n_1, \ldots, n_N\}$, edge set $\mathcal{E} = \{(n_i, n_j) \in \mathcal{V} \times \mathcal{V}\}$, and adjacency matrix $\Lambda = [a_{ij}] \in \mathbb{R}^{N\times N}$. $(n_i, n_j) \in \mathcal{E}$ means the message flow from agent i to j over the time. The element a_{ij} is a nonnegative constant satisfying a_{ij}=1, if $n_j, n_i \in \mathcal{E}$, and $a_{ij} = 0$ if $n_j, n_i \notin \mathcal{E}$, or $i = j$. Define the Laplacian matrix as $\mathcal{L} = [l_{ij}] \in \mathbb{R}^{N\times N}$ with $l_{ii} = \sum_{j=1, j\neq i}^{N} a_{ij}$ and $l_{ij} = -a_{ij},\ i \neq j$. A directed communication path from node n_1 to n_k is a sequence of ordered edges $(n_1, n_2), (n_2, n_3), \ldots, (n_{k-1}, n_k)$.

B. Problem Formulation

The uncertain nonlinear systems consist of N formation leaders, labeled as $\mathcal{V}_l = \{1, \ldots, N\}$, and M followers, labeled as $\mathcal{V}_f = \{N+1, \ldots, N+M\}$. The dynamics model of the ith agent is denoted as

$$\begin{cases} \dot{\mathbf{X}}_{i,p} = \mathbf{X}_{i,v}, \\ \dot{\mathbf{X}}_{i,v} = \mathbf{f}_{i,v}(\mathbf{X}_{i,v}) + \mathbf{u}_i + \tau_i \\ \mathbf{Y}_i = \mathbf{X}_{i,p} \end{cases} \tag{1}$$

where $i \in \mathcal{V}_f \cup \mathcal{V}_l$, $\mathbf{X}_{i,p} \in \mathbb{R}^3$, $\mathbf{X}_{i,v} \in \mathbb{R}^3$ are position and velocity components of ith agent. $\mathbf{u}_i \in \mathbb{R}^3$ and $\mathbf{Y}_i \in \mathbb{R}^3$ represent the control input and output, respectively. $\mathbf{f}_{i,v}(\mathbf{X}_{i,v}) \in \mathbb{R}^3$ denotes the unknown nonlinear term associated with the velocity vector, and the vector of unknown disturbances is given by $\tau_i \in \mathbb{R}^3$. Consider a virtual leader, labeled as 0, with the following dynamics:

$$\dot{\mathbf{X}}_{0,p} = \mathbf{X}_{0,v}, \quad \dot{\mathbf{X}}_{0,v} = \mathbf{u}_0 \tag{2}$$

where $\mathbf{X}_{0,p} \in \mathbb{R}^3$, $\mathbf{X}_{0,v} \in \mathbb{R}^3$ are position and velocity of the virtual leader and $\mathbf{u}_0 \in \mathbb{R}^3$ is the control input.

Note that only a subset of the formation leaders can obtain the position and velocity state of virtual leader. The Laplacian matrix can also be expressed as follows:

$$\mathcal{L} = \begin{bmatrix} 0 & 0_{1\times N} & 0_{1\times M} \\ \mathcal{L}_{L2} & \mathcal{L}_{L1} & 0_{N\times M} \\ 0_{M\times 1} & \mathcal{L}_{F2} & \mathcal{L}_{F1} \end{bmatrix}$$

where $\mathcal{L}_{L1} \in \mathbb{R}^{N\times N}$ denotes communication links among formation leaders. $\mathcal{L}_{F1} \in \mathbb{R}^{M\times M}$ describes the communication topology among followers. $\mathcal{L}_{F2} \in \mathbb{R}^{M\times N}$ means the information interaction between the followers and formation leaders. Define $\mathcal{L}_{L2} = [-a_{10}, -a_{20}, \ldots, -a_{N0}]^T$. $a_{i0} > 0$ $(i = 1, \ldots, N)$ represents the formation leader can access to the desired command of virtual leader indexed by $j = 0$, and $a_{i0} = 0$ otherwise.

Remark 1: A class of physical systems can be modeled as the uncertain nonlinear systems (1), such as inner-loop/outer-loop subsystem of quadrotor [1], [32], rigid spacecraft [33], etc.

Assumption 1 [1]: Regarding each follower, there exists at least one directed communication path from formation leaders. Additionally, the communication link among the leaders is undirected and at least one of the formation leaders maintains connected with the virtual leader.

Lemma 1 [1]: Supposed that Assumption 1 holds, each element of $-\mathcal{L}_{F1}^{-1}\mathcal{L}_{F2}$, $-\mathcal{L}_{L1}^{-1}\mathcal{L}_{L2}$ is nonnegative, and the sum of each row for $-\mathcal{L}_{F1}^{-1}\mathcal{L}_{F2}$, $-\mathcal{L}_{L1}^{-1}\mathcal{L}_{L2}$ is one. Moreover, $\mathcal{L}_{L1} \in \mathbb{R}^{N\times N}$ is symmetric positive definite.

Assumption 2 [32]: Denote the lumped disturbances as $\varphi_i = [\varphi_{i,1}, \varphi_{i,2}, \varphi_{i,3}]^T = \mathbf{f}_{i,v}(\mathbf{X}_{i,v}) + \tau_i$, which satisfies $|\dot{\varphi}_{i,\eta}| \leq L_{i,v}$ $(\eta = 1, 2, 3)$ with $L_{i,v}$ being a positive constant.

Remark 2: Considering the fact that state-related nonlinear terms and external disturbances are energy restrained in reality, the existence of a constant bound for lumped disturbance by Assumption 2 is fairly common in rejection disturbance control [2], [32].

1) Control Objective: For MASs (1) and (2) with resources and response time constricts, a hierarchical DFTFFC strategy is developed, which consists of distributed fixed-time estimators involving event-triggered communications and backstepping controller employing intermittent output feedback, such that

1) The formation leaders shape a predefined geometrical configuration centered on the virtual leader and each follower reaches the convex hull within a preset time.
2) The utilization of system resources is optimized and Zeno behavior is avoided.

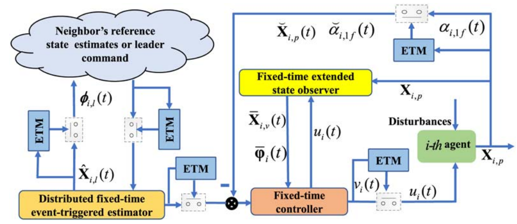

Fig. 1. Control structure of agent $i \in \mathcal{V}_f \cup \mathcal{V}_l$.

The following lemmas are required for the stability analysis.

Lemma 2 [25]: Assume that there exists a Lyapunov function satisfying the condition

$$\dot{V}(x(t)) \leq -(\alpha V^p(x(t)) + \beta V^q(x(t)))^k$$

with $\alpha > 0, \beta > 0, pk \in (0,1)$ and $qk \in (1, \infty)$. Then we can derive that system (1) can be stabilized within fixed time regardless of initial states, and the setting time equation is

$$T \leq T_{\max} = \frac{1}{\alpha^k(1-pk)} + \frac{1}{\beta^k(qk-1)}.$$

Additionally, assume that

$$\dot{V}(x(t)) \leq -(\alpha V^p(x(t)) + \beta V^q(x(t)))^k + \Phi$$

with $\Phi > 0$. The system (1) is practical fixed-time stable, which means that the error signal will remain in a small region around zero when t approaches $T_{\max}$.

Lemma 3 [26]: Let $\zeta_1, \zeta_2 \ldots, \zeta_N \geq 0$, we have

$$\sum_{i=1}^{N} {\zeta_i}^p \geq \left(\sum_{i=1}^{N} \zeta_i\right)^p, \quad 0 \leq p \leq 1 \tag{3}$$

$$N^{1-q}\left(\sum_{i=1}^{N} \zeta_i\right)^q \leq \sum_{i=1}^{N} \zeta_i^q \leq \left(\sum_{i=1}^{N} \zeta_i\right)^q, \quad 1 < q \leq \infty. \tag{4}$$

III. Fixed-Time Event-Triggered Formation Control Design for Formation Leaders

This section consists of three parts: 1) DFTETEs are constructed to extract the desired location and velocity of leaders in the formation within a predefined time, while reducing unnecessary communication among neighbors. 2) the fixed-time extended state observer (FTESO) is designed to exactly reconstruct the unmeasured velocity and lumped disturbances of nonlinear uncertain systems (1). 3) a fixed-time backstepping controller based on intermittent output is proposed to reduce the unnecessary controller update and communication occupation between the sensor-to-controller channel and controller-to-actuator channel. The overall architecture of the controller is depicted in Fig. 1.

To improve readability, the notations are listed in Table I.

TABLE I
NOMENCLATURE

$\hat{\mathbf{X}}_{i,p}$	Position estimation of desired state	$\mathbf{s}_{i,p}$	Tracking error of position loop
$\hat{\mathbf{X}}_{i,v}$	Velocity estimation of desired state	$\mathbf{s}_{i,v}$	Tracking error of velocity loop
$\bar{\mathbf{X}}_{i,p}$	Position estimation of system state	$\alpha_{i,1}$	Virtual control
$\bar{\mathbf{X}}_{i,v}$	Velocity estimation of system state	$\alpha_{i,1f}$	Filter output
$\bar{\varphi}_i$	Lumped disturbance estimation	$\mathbf{u}_i$	Actual control input

A. Distributed Fixed-Time Event-Triggered Estimator

As the desired formation configuration and the virtual leaders' information is only accessible to a subset of formation leaders, we construct a DFTETE for formation leaders

$$\begin{aligned}
\dot{\hat{x}}_{i,l\eta} &= -\beta_1^{-\frac{1}{\gamma_2}}\mathrm{sig}^{\frac{1}{\gamma_2}}(y_{il\eta}+\alpha_1^*\mathrm{sig}^{\gamma_1}(y_{il\eta}))-\lambda_1\mathrm{sign}(y_{il\eta})(l=p,v)\\
y_{\mathrm{ip}\eta} &= \sum_{j=0}^{N} a_{ij}\left(\hat{x}_{i,p\eta}(t)-\hat{x}_{j,p\eta}(t_{j,p\eta}^{k'})-\delta_{ij,\eta}\right)\\
y_{iv\eta} &= \sum_{j=0}^{N} a_{ij}(\hat{x}_{i,v\eta}(t)-\hat{x}_{j,v\eta}(t_{j,v\eta}^{k'}))
\end{aligned} \tag{5}$$

where $\hat{x}_{i,l\eta}\in\hat{\mathbf{X}}_{i,l}$ $(i\in\mathcal{V}_l,\eta=1,2,3)$ stand for the estimations of desired state for formation leaders. $\hat{x}_{0,p\eta}=x_{0,p\eta}$, $\hat{x}_{0,v\eta}=\dot{x}_{0,p\eta}$ are position and velocity of the virtual leader. $\delta_{ij}=\delta_i-\delta_j=[\delta_{ij,1},\delta_{ij,2},\delta_{ij,3}]^T$ with $\delta_{ij,\eta}$ represents the relative distance between neighbors and $\delta_{i,\eta},\delta_{j,\eta}$ are two constants to be designed. $\alpha_1^*,\beta_1,\lambda_1$ are positive estimator gains. γ_1,γ_2 contribute to the parameters of power exponent satisfying $2\gamma_2>\gamma_1>\gamma_2, 1<\gamma_2<2$. $t_{j,l\eta}^{k'}=\mathrm{argmin}_{s\in N^+ t_{j,l\eta}^{k'}>t_{j,l\eta}^{s}}(t-t_{j,l\eta}^{s})$ means the latest triggered time of neighbor agent j. Moreover, define $\mathrm{sig}^{\gamma}(\cdot)=|\cdot|^{\gamma}\mathrm{sign}(\cdot)$ with $\mathrm{sign}(\cdot)$ being the signum function and $\gamma>0$.

The triggering condition for DFTETE is designed as

$$\begin{aligned}
&\dot{\phi}_{i,l\eta}(t)=\dot{\hat{x}}_{i,l\eta}(t_{i,l\eta}^k),\forall t\in[t_{i,l\eta}^k,t_{i,l\eta}^{k+1}), e_{i,l\eta}^1(t)=\dot{\phi}_{i,l\eta}(t)-\dot{\hat{x}}_{i,l\eta}(t)\\
&\phi_{i,l\eta}(t)=\hat{x}_{i,l\eta}(t_{i,l\eta}^k),\forall t\in[t_{i,l\eta}^k,t_{i,l\eta}^{k+1}), e_{i,l\eta}^2(t)=\phi_{i,l\eta}(t)-\hat{x}_{i,l\eta}(t)\\
&t_{i,l\eta}^{k+1}=\inf\{t>t_{i,l\eta}^k:\left|\mathbf{e}_{i,l\eta}^1(t)\right|\geq\xi_{i,l\eta}^1 or \left|\mathbf{e}_{i,l\eta}^2(t)\right|\geq\xi_{i,l\eta}^2\}
\end{aligned} \tag{6}$$

where $\dot{\phi}_{i,l\eta}(t)$ and $\phi_{i,l\eta}(t)$ are generated by $\dot{\hat{x}}_{i,l\eta}(t)$ and $\hat{x}_{i,l\eta}(t)$ when $t=t_{i,l\eta}^k$. $t_{i,l\eta}^k$ represents the triggered instant of $\hat{x}_{i,l\eta}(t_{i,l\eta}^k)$. $\hat{x}_{i,l\eta}(t_{i,l\eta}^k)$ will be updated as $\hat{x}_{i,l\eta}(t_{i,l\eta}^{k+1})$ only when event-triggered boundary (6) is violated. $e_{i,l\eta}^1, e_{i,l\eta}^2$ are the measurement errors. $\xi_{i,l\eta}^1$ and $\xi_{i,l\eta}^2$ represent the constant triggering thresholds.

Besides, given the following vectors:

$Q_L=[\mathbf{X}_{1,p}^T,\mathbf{X}_{2,p}^T,\ldots,\mathbf{X}_{N,p}^T]^T, Q_0=[\mathbf{X}_{0,p}^T,\mathbf{X}_{0,p}^T,\ldots,\mathbf{X}_{0,p}^T]^T$ to denote the position sets of formation leaders and virtual leader.

Theorem 1: Under Assumption 1, if DFTETE (5) and ETM (6) are adopted for the considered formation leaders, the estimation errors can converge to a small region around zero within a fixed time and the Zeno behavior is excluded.

Proof: The proof process of Theorem 1 is shown in Appendix A.

Remark 3: The difficulty of proving the stability of the DFTETEs lies in the coupling between the triggering errors and the nonlinearity of the distributed fixed-time observer.

B. Fixed-Time Extended State Observer

A FTESO based on measurement states is developed for nonlinear system (1)

$$\begin{cases}
\dot{\bar{\mathbf{X}}}_{i,p}=\bar{\mathbf{X}}_{i,v}+\ell_{i,1}\theta(t)\mathrm{sig}(\mathbf{X}_{i,p}-\bar{\mathbf{X}}_{i,p})^{2/3}\\
\qquad+\kappa_{i,1}(1-\theta(t))\mathrm{sig}(\mathbf{X}_{i,p}-\bar{\mathbf{X}}_{i,p})^{(3+m)/3}\\
\dot{\bar{\mathbf{X}}}_{i,v}=\bar{\varphi}_i+u_i+\ell_{i,2}\theta(t)\mathrm{sig}(\mathbf{X}_{i,p}-\bar{\mathbf{X}}_{i,p})^{1/3}\\
\qquad+\kappa_{i,2}(1-\theta(t))\mathrm{sig}(\mathbf{X}_{i,p}-\bar{\mathbf{X}}_{i,p})^{(3+2m)/3}\\
\dot{\bar{\varphi}}_i=\ell_{i,3}\theta(t)\mathrm{sign}(\mathbf{X}_{i,p}-\bar{\mathbf{X}}_{i,p})\\
\qquad+\kappa_{i,3}(1-\theta(t))\mathrm{sig}(\mathbf{X}_{i,p}-\bar{\mathbf{X}}_{i,p})^{1+m}
\end{cases} \tag{7}$$

where the estimation of system states $\mathbf{X}_{i,p},\mathbf{X}_{i,v}$ are denoted as $\bar{\mathbf{X}}_{i,p}$ and $\bar{\mathbf{X}}_{i,v}$ respectively. In addition, $\bar{\varphi}_i$ is the estimation of lumped disturbances φ_i.

In order to ensure fixed-time stabilization of the designed FTESO, the observer parameters satisfy $\ell_{i,1}=2L_{i,v}^{(1/3)},\ell_{i,2}=1.5L_{i,v}^{(2/3)}$ and $\ell_{i,3}=1.1L_{i,v}$; $m>0$ is a small positive constant and the parameter matrix $\mathbf{A}_i=[-\kappa_{i,1},1,0;-\kappa_{i,2},0,1;-\kappa_{i,3},0,0]$ is Hurwitz; the switching scheme of error polynomials is devised by $\theta(t)=0$, if $t\leq T_u$, conversely $\theta(t)=1$, where T_u is a positive real number.

The observer errors are $\mathbf{Z}_{i,p}=\mathbf{X}_{i,p}-\bar{\mathbf{X}}_{i,p}$, $\mathbf{Z}_{i,v}=\mathbf{X}_{i,v}-\bar{\mathbf{X}}_{i,v}$, $\mathbf{Z}_{i,\varphi}=\varphi_i-\bar{\varphi}_i$. Then, the error dynamics can be given as follows:

$$\begin{cases}
\dot{\mathbf{Z}}_{i.p}=\mathbf{Z}_{i.v}-\ell_{i,1}\theta(t){\mathbf{Z}_{i.p}}^{2/3}-\kappa_{i,1}(1-\theta(t)){\mathbf{Z}_{i.p}}^{(3+m)/3}\\
\dot{\mathbf{Z}}_{i.v}=\mathbf{Z}_{i,\varphi}-\ell_{i,2}\theta(t){\mathbf{Z}_{i.p}}^{1/3}-\kappa_{i,2}(1-\theta(t)){\mathbf{Z}_{i.p}}^{(3+2m)/3}\\
\dot{\mathbf{Z}}_{i,\varphi}=\dot{\varphi}_i-\ell_{i,3}\theta(t)\mathrm{sign}(\mathbf{Z}_{i.p})-\kappa_{i,3}(1-\theta(t)){\mathbf{Z}_{i.p}}^{1+m}.
\end{cases} \tag{8}$$

Note that the error dynamics (8) of developed FTESO are inspired by fixed-time exact differentiator [34], [35], the same analysis process is omitted. As a result, we can derive that observer errors $\mathbf{Z}_{i,p},\mathbf{Z}_{i,v},\mathbf{Z}_{i,\varphi}$ decay to origin within bounded convergence time, and the reaching time is only dependent on observer parameters.

Remark 4: The FTESO introduces two polynomial feedback with different powers, which work alternately based on the switching function $\theta(t)$. For the transient phase when observer error generally satisfies $\mathbf{Z}_{i,p}\gg 1$, polynomials with powers greater than one can provide faster convergence. For the steady phase, polynomials with fractional powers ensure the finite-time stabilization of error systems.

C. Formation Controller Design of Formation Leaders With Intermittent Output

After acquiring desired states of formation leaders, distributed formation control problem of MASs can be turned into a local tracking control problem of agents. Then, a robust fixed-time backstepping control protocol using a two-layer ETMs is proposed to steer the formation leaders into the prespecified formation pattern within bounded reaching time.

The design process is divided into two parts. First, we present the fixed-time controller without considering ETMs.

Step 1: The tracking error of position loop is denoted as

$$\mathbf{s}_{i,p} = \mathbf{X}_{i,p} - \phi_{i,p}. \tag{9}$$

Differentiating (9) yields

$$\dot{\mathbf{s}}_{i,p} = \dot{\mathbf{X}}_{i,p} - \dot{\phi}_{i,p}. \tag{10}$$

For sake of stabilizing the error dynamic (10), a fixed-time virtual control law is design as

$$\alpha_{i,1} = \phi_{i,v} - k_{i,p}[\mathbf{s}_{i,p} + \text{sig}^p(\mathbf{s}_{i,p}) + \text{sig}^q(\mathbf{s}_{i,p})] + d_1 \tag{11}$$

where $k_{i,p}$ is the virtual controller parameter, power exponents satisfy $0 < p < 1$ and $1 < q < 2$, and $d_1 = \dot{\phi}_{i,p}(t) - \phi_{i,v}(t)$ is utilized to offset the uncertainties induced by event-triggered.

Then, we introduce a first-order low-pass filter to avoid the differential explosion issue of $\alpha_{i,1}$

$$\kappa\dot{\alpha}_{i,1f} = \alpha_{i,1} - \alpha_{i,1f}, \alpha_{i,1f}(0) = \alpha_{i,1}(0) \tag{12}$$

where $\kappa > 0$ is a filter gain.

Define the filter error as

$$\Delta_i = \alpha_{i,1f} - \alpha_{i,1}. \tag{13}$$

Step 2: The tracking error and its differential of velocity loop are expressed as

$$\begin{cases} \mathbf{s}_{i,v} = \bar{\mathbf{X}}_{i,v} - \alpha_{i,1f} \\ \dot{\mathbf{s}}_{i,v} = u_i + \varphi_i - \dot{\alpha}_{i,1f} - \dot{\mathbf{Z}}_{i,v}. \end{cases} \tag{14}$$

Subsequently, with the aid of backstepping technique, the fixed-time controller is construct as

$$u_i = -\mathbf{s}_{i,p} - k_{i,v}[\mathbf{s}_{i,v} + \text{sig}^p(\mathbf{s}_{i,v}) + \text{sig}^q(\mathbf{s}_{i,v})] - \bar{\varphi}_{i,v} + \dot{\alpha}_{i,1f} \tag{15}$$

where $k_{i,v}$ is the controller parameter.

The preceding discussions fulfill the first part of the controller design.

Second, the ETMs are introduced to govern the information flow of output states

$$\begin{aligned} &\breve{x}_{i,p\eta}(t) = x_{i,p\eta}(t^{\sigma}_{i,p\eta}), \quad t \in [t^{\sigma}_{i,p\eta}, t^{\sigma+1}_{i,p\eta}) \\ &t^{\sigma+1}_{i,p\eta} = \inf\{t > t^{\sigma}_{i,p\eta} : \left|\breve{x}_{i,p\eta}(t) - x_{i,p\eta}(t)\right| \geq c_{i,1}\left|\breve{s}_{i,p\eta}\right|\} \end{aligned} \tag{16}$$

where $\breve{x}_{i,p\eta}(t)$ is the triggered state of $x_{i,p\eta}(t)$ within the time internal $t \in [t^{\sigma}_{i,p\eta}, t^{\sigma+1}_{i,p\eta})$, This state will be updated when $t = t^{\sigma+1}_{i,p\eta}$. Additionally, $c_{i,1}\left|\breve{s}_{i,p\eta}\right|$ is a designed positive threshold boundary, with $c_{i,1}$ being a positive constant and $\breve{s}_{i,p\eta} \in \breve{\mathbf{s}}_{i,p}$ is given in the following equation:

From (9) and (16), the tracking errors of position loop and virtual control laws are rewritten as

$$\begin{aligned} &\breve{\mathbf{s}}_{i,p} = \breve{\mathbf{X}}_{i,p} - \phi_{i,p} \\ &\bar{\alpha}_{i,1} = \phi_{i,v} - k_{i,p}[\breve{\mathbf{s}}_{i,p} + \text{sig}^p(\breve{\mathbf{s}}_{i,p}) + \text{sig}^q(\breve{\mathbf{s}}_{i,p})] + d_1. \end{aligned} \tag{17}$$

To solve the nondifferentiable problem of $\bar{\alpha}_{i,1}$ caused by intermittent output $\breve{\mathbf{X}}_{i,p}$, a new triggered filter output $\breve{\alpha}_{i,1f}$ is developed

$$\begin{aligned} &\breve{\alpha}_{i,1f\eta}(t) = \alpha_{i,1f\eta}(t^{\sigma'}_{i,\alpha\eta}) \quad t \in [t^{\sigma'}_{i,\alpha\eta,}, t^{\sigma'+1}_{i,\alpha\eta}) \\ &t^{\sigma'+1}_{i,\alpha\eta} = \inf\{t > t^{\sigma'}_{i,\alpha\eta} : \left|\breve{\alpha}_{i,1f\eta}(t) - \alpha_{i,1f\eta}(t)\right| \geq c'_{i,1}\left|\breve{\mathbf{s}}_{i,v\eta}\right|\} \end{aligned} \tag{18}$$

where $\breve{\alpha}_{i,1f\eta}(t)$ is the triggered state of $\alpha_{i,1f\eta}(t^{\sigma'}_{i,\alpha\eta})$ within the time internal $t \in [t^{\sigma'}_{i,p\eta}, t^{\sigma'+1}_{i,p\eta})$, This state will be updated when $t = t^{\sigma'+1}_{i,p\eta}$. Additionally, $c'_{i,1}\left|\breve{\mathbf{s}}_{i,v\eta}\right|$ is a design threshold with $c'_{i,1}$ being a positive constant and $\breve{\mathbf{s}}_{i,v\eta} \in \breve{\mathbf{s}}_{i,v}$ is given in (20).

Then, the tracking error of velocity loop under intermittent output is given as

$$\breve{\mathbf{s}}_{i,v} = \bar{\mathbf{X}}_{i,v} - \breve{\alpha}_{i,1f}. \tag{19}$$

Following (19), we employ the discontinuous errors to redesign actual control algorithm reducing the frequency of controller update

$$\begin{aligned} v_i = &-\breve{\mathbf{s}}_{i,p} - k_{i,v}[\breve{\mathbf{s}}_{i,v} + \text{sig}^p(\breve{\mathbf{s}}_{i,v}) + \text{sig}^q(\breve{\mathbf{s}}_{i,v})] \\ &- \bar{\varphi}_i + (\bar{\alpha}_{i,1} - \breve{\alpha}_{i,1f})/\kappa. \end{aligned} \tag{20}$$

Afterward, in order to reduce the communication load on the controller-to-actuator channel, we proceed to design a second-layer event trigger tailored for discontinuous control input as

$$\begin{aligned} &u_{i,\eta}(t) = v_{i,\eta}(t^{\sigma'}_{i,u}) \quad t \in [t^{\sigma'}_{i,u}, t^{\sigma'+1}_{i,u}) \\ &t^{\sigma'+1}_{i,u\eta} = \inf\{t > t^{\sigma'}_{i,u} : |u_{i,\eta}(t) - v_{i,\eta}(t)| \geq c_{i,2}\left|\breve{s}_{i,v\eta}\right|\} \end{aligned} \tag{21}$$

where $u_{i,\eta}$ is the triggered state of $v_{i,\eta}$ within the time internal $t \in [t^{\sigma'}_{i,u}, t^{\sigma'+1}_{i,u})$, which will be assigned as $v(t^{\sigma'+1}_{i,u})$ until the preset threshold is satisfied at time instant $t^{\sigma'+1}_{i,u}$. Besides, $c_{i,2}\left|\breve{s}_{i,v\eta}\right|$ is the threshold boundary with $c_{i,2}$ being a positive constant and $\breve{\mathbf{s}}_{i,v\eta} \in \breve{\mathbf{s}}_{i,v}$ is given in (20).

Remark 5: Note that the differentiation of virtual control law (19) is undefined because it involves the intermittent output $\breve{x}_{i,p\eta}$. With the aid of a first-order low-pass filter, the differential of $\bar{\alpha}_{i,1}$ can be replaced by the filter output so that the problem is solved. However, during the stability analysis, the problem above arises again in the differentiation of filtering error. Herein, the key step of this method involves the design of an event-triggered mechanism for filter output $\alpha_{i,1f}$, utilizing the triggering filter output $(\bar{\alpha}_{i,1} - \breve{\alpha}_{i,1f\eta})/\kappa$ instead of $(\alpha_{i,1} - \alpha_{i,1f\eta})/\kappa$. As a result, the error terms (9) and (14) can be rewritten as error dynamics (30) and (31) comprising measurement errors (16), (18), and (21), filter errors (13), and tracking error polynomials. The nondifferentiable issues existing in the fixed-time stability analysis are addressed.

Theorem 2: Considering the formation leader systems (1), suppose the DFTETE (5) satisfies Theorem 1. Given any

$\Gamma_1 > 0$, for all initial conditions satisfying $\mathbf{s}_{i,p}^T\mathbf{s}_{i,p} + \mathbf{s}_{i,v}^T\mathbf{s}_{i,v} + \mathbf{Z}_{i,p}^T\mathbf{Z}_{i,p} + \mathbf{Z}_{i,v}^T\mathbf{Z}_{i,v} + \mathbf{Z}_{i,\varphi}^T\mathbf{Z}_{i,\varphi} + {\Delta_i}^T\Delta_i \leq 2\Gamma_1, i \in \mathcal{V}_l$, the tracking errors $\mathbf{s}_{i,p}, \mathbf{s}_{i,v}$ will semiglobally converge to a compact set around the origin under the FTESO (7), the fixed-time event-triggered virtual control laws (17) and the fixed-time event-triggered controller (20). The convergence time satisfies $t \leq T_e + T_c$. Meanwhile, the Zeno phenomenon can be excluded.

Proof: The proof process of Theorem 2 is shown in Appendix B.

IV. Fixed-Time Event-Triggered Containment Control Design for Followers

In this section, the distributed fixed-time event-triggered containment control scheme is proposed for the followers. Similarly, a DFTETE is proposed for followers to estimate the convex hull formed by network-linked multiple formation leaders. Then, a fixed-time backstepping controller is proposed for followers with discontinuous output.

A. Distributed Fixed-Time Event-Triggered Estimator

From Theorem 2, it can be derived that all states of the formation leaders are bounded. We propose a DFTETE for ith followers in the form of

$$\begin{aligned}\dot{\hat{x}}_{i,l\eta} &= -\beta_2^{-\frac{1}{\gamma_4}}\mathrm{sig}^{\frac{1}{\gamma_4}}(y_{il\eta} + \alpha_2^*\mathrm{sig}^{\gamma_3}(y_{il\eta})) - \lambda_2\mathrm{sign}(y_{il\eta})(l{=}p,v)\\ y_{\mathrm{ip}\eta} &= \sum_{j=1}^{N+M} a_{ij}(\hat{x}_{i,p\eta}(t) - \hat{x}_{j,p\eta}(t_{j,p\eta}^{k'}))\\ y_{iv\eta} &= \sum_{j=1}^{N+M} a_{ij}(\hat{x}_{i,v\eta}(t) - \hat{x}_{j,v\eta}(t_{j,v\eta}^{k'}))\end{aligned} \tag{22}$$

where $\hat{x}_{i,l\eta} \in \hat{\mathbf{X}}_{i,l}$ $(i \in \mathcal{V}_f, \eta = 1, 2, 3)$ stand for the estimations of the weighed mean of multiple formation leaders' positions and velocities for followers. $\alpha_2^*, \beta_2, \lambda_2$ are positive estimator gains. γ_3, γ_4 contribute to the parameters of power exponent satisfying $2\gamma_4 > \gamma_3 > \gamma_4, 1 < \gamma_4 < 2$.

The triggering condition of DFTETE (22) is obtained by referring to (6). Besides, given the following vectors:

$$\begin{aligned}Q_F &= [\mathbf{X}_{N+1,p}^T, \mathbf{X}_{N+2,p}^T, \ldots, \mathbf{X}_{N+M,p}^T]^T\\ H_F &= [h_{N+1,p}, h_{N+2,p}, \ldots, h_{N+M,p}]^T = -(\mathcal{L}_{F1}^{-1}\mathcal{L}_{F2} \otimes I_3)Q_L\end{aligned}$$

to denote the sets of follower positions and weighted mean of formation leader positions.

Theorem 3: Under Assumption 1, if the DFTETE (22) is designed for the considered followers, the estimation errors can converge to a small region around zero within a fixed time and the non-Zeno behavior is proven.

Proof: Consider the estimator errors as $\tilde{z}_{i,p\eta} = \hat{x}_{i,p\eta} - h_{i,p\eta}$, $\tilde{z}_{i,v\eta} = \hat{x}_{i,v\eta} - \dot{h}_{i,p\eta}$ and given the vector form $\tilde{\mathbf{z}}_p = [\tilde{z}_{N+1,p}^T, \ldots, \tilde{z}_{N+M,p}^T]^T$, $\tilde{\mathbf{z}}_v = [z_{N+1,v}^T, \ldots, z_{N+M+m,v}^T]^T$.

Design the Lyapunov function for the DFTETE (22)

$$\mathbf{V}_4 = \frac{1}{2}\tilde{\mathbf{z}}_v^T(\mathcal{L}_{F1} \otimes \mathbf{I}_3)\tilde{\mathbf{z}}_v, \quad \mathbf{V}_5 = \frac{1}{2}\tilde{\mathbf{z}}_p^T(\mathcal{L}_{F1} \otimes \mathbf{I}_3)\tilde{\mathbf{z}}_p.$$

With the aid of a similar analysis to Theorem 1, the practical fixed-time stabilization for estimation errors can be proven, while avoiding the Zeno behavior. The convergence time is given as

$$T_e' \leq \frac{1}{(\frac{\tilde{\varpi}_1}{\beta_2}(1-\psi))^{\frac{1}{\gamma_4}}(1 - \frac{\gamma_4+1}{2}\frac{1}{\gamma_4})} + \frac{1}{(\frac{\alpha_2\tilde{\varpi}_2}{\beta_2})^{\frac{1}{\gamma_4}}(\frac{\gamma_4+\gamma_3}{2}\frac{1}{\gamma_4} - 1)}$$

where

$$\begin{aligned}\tilde{\varpi}_1 &= (3M)^{\frac{1-\gamma_4}{2}}(2\lambda_{\min}(\mathcal{L}_{F1}))^{\frac{1+\gamma_4}{2}},\\ \tilde{\varpi}_2 &= (3M)^{\frac{2-\gamma_3-\gamma_4}{2}}(2\lambda_{\min}(\mathcal{L}_{F1}))^{\frac{\gamma_3+\gamma_4}{2}}.\end{aligned}$$

B. Containment Controller Design of Followers With Intermittent Output

A part of the detail of the controller design is omitted due to the similarity with part C of Section III.

The virtual control law and actual containment control algorithm are developed as

$$\begin{aligned}\bar{\alpha}_{i,1} &= \phi_{i,v} - k_{i,p}[\breve{\mathbf{s}}_{i,p} + \mathrm{sig}^p(\breve{\mathbf{s}}_{i,p}) + \mathrm{sig}^q(\breve{\mathbf{s}}_{i,p})] + \tilde{d}_1,\\ v_i &= -\breve{\mathbf{s}}_{i,p} - k_{i,v}[\breve{\mathbf{s}}_{i,v} + \mathrm{sig}^p(\breve{\mathbf{s}}_{i,v}) + \mathrm{sig}^q(\breve{\mathbf{s}}_{i,v})]\\ &\quad - \bar{\varphi}_i + (\bar{\alpha}_{i,1} - \breve{\alpha}_{i,1f})/\kappa\end{aligned} \tag{23}$$

where $i \in \mathcal{V}_f$; $k_{i,p}$ and $k_{i,v}$ are the controller gains; $\tilde{d}_1 = \dot{\phi}_{i,p} - \phi_{i,v}$ is the compensation term.

Next, the triggering threshold for the actual controller is presented as

$$\begin{cases} u_{i,\eta}(t) = v_{i,\eta}(t_{i,u}^{\sigma'}), & t \in [t_{i,u}^{\sigma'}, t_{i,u}^{\sigma'+1})\\ t_{i,u\eta}^{\sigma'+1} = \inf\{t > t_{i,u}^{\sigma'} : |u_{i,\eta}(t) - v_{i,\eta}(t)| \geq \tilde{c}_{i,2}\left|\breve{s}_{i,v\eta}\right|\} \end{cases} \tag{24}$$

where $u_{i,\eta} (i \in \mathcal{V}_f)$ is the triggered state of $v_{i,\eta}$ within the time internal $t \in [t_{i,u}^{\sigma'}, t_{i,u}^{\sigma'+1})$, which will be assigned as $v(t_{i,u}^{\sigma'+1})$ until the preset threshold is satisfied at time instant $t_{i,u}^{\sigma'+1}$. Besides, $\tilde{c}_{i,2}\left|\breve{s}_{i,v\eta}\right|$ is the threshold boundary with $\tilde{c}_{i,2}$ being a positive constant and $\breve{\mathbf{s}}_{i,v\eta} \in \breve{\mathbf{s}}_{i,v}$ can be determined by referring to (20).

Theorem 4: Considering the follower systems (1), suppose the DFTETE (22) satisfies Theorem 3. Given any $\Gamma_2 > 0$, for all initial conditions satisfying $\mathbf{s}_{i,p}^T\mathbf{s}_{i,p} + \mathbf{s}_{i,v}^T\mathbf{s}_{i,v} + \mathbf{Z}_{i,p}^T\mathbf{Z}_{i,p} + \mathbf{Z}_{i,v}^T\mathbf{Z}_{i,v} + \mathbf{Z}_{i,\varphi}^T\mathbf{Z}_{i,\varphi} + {\Delta_i}^T\Delta_i \leq 2\Gamma_2, i \in \mathcal{V}_f$, the tracking errors $\mathbf{s}_{i,p}, \mathbf{s}_{i,v}$ will semiglobally converge to a compact set around the origin under the FTESO (7) and the fixed-time event-triggered virtual control laws (23). The convergence time satisfies $t \leq T_e' + T_c'$. Meanwhile, the Zeno phenomenon can be excluded.

Proof: The Lyapunov function $V_{6,i}$ is selected to be identical to $V_{3,i}$. Similar proof can refer to (30)–(38), we can get

$$\dot{V}_{6,i} \leq -2^{\frac{1-p}{2}}\tilde{\omega}\tilde{\rho}_{i,2}{V_{6,i}}^{(p+1)/2} - 2^{\frac{1-q}{2}}\tilde{\omega}\tilde{\rho}_{i,3}{V_{3,i}}^{(q+1)/2} + \tilde{B}$$

where $\tilde{\rho}_{i,2} > 0, \tilde{\rho}_{i,3} > 0, \tilde{\omega} > 0$, and the constant bound $\tilde{B} > 0$.

The upper bound of convergence time is expressed as

$$T_c^{'} = \frac{2}{\sum_{i=1}^{N} 2^{\frac{1-p}{2}} \tilde{\omega}\tilde{\rho}_{i,2}(1-p)} + \frac{2}{\sum_{i=1}^{N} 2^{\frac{1-q}{2}} \tilde{\omega}\tilde{\rho}_{i,3}(q-1)}. \tag{25}$$

As a result, we can conclude that the error signals $\mathbf{s}_{i,p}, \mathbf{s}_{i,v} (i \in \mathcal{V}_f \cup \mathcal{V}_l)$ converge to a compact set after $t = T_e + T_c + T_e^{'} + T_c^{'}$. which implies that the practical fixed-time stabilization is realized for uncertain nonlinear MASs with intermittent output, connectivity, and actuation. The objectives of the formation leader to shape a predefined geometrical configuration and each follower to reach the convex hull are accomplished simultaneously. Moreover, the Zeno behavior is avoided. ■

Remark 6: The highlight of the research in practice is to minimize resource cost by intermittent output, connectivity, and actuation, while still ensuring a bounded settling time without retuning parameters across various initial conditions. In theory, the parameter selection range is determined by Theorem 2. Notice that a larger power q and a smaller power p can expedite convergence, but this may lead to input saturation behavior. Hence, a tradeoff between achieving optimal system performance and maintaining a tolerable control force needs to consider.

Remark 7: Compared with the distributed observer-based results [14], in which the asymptotic tracking can be ensured, the developed strategy ensures the practical fixed-time stability of closed-loop systems. Compared with fixed-time observer-based consensus control schemes [18], this article provides a solution to the DFTFCC problem for uncertain nonlinear systems, of which the solution for consensus issue can be regarded as a special case. Compared with the ETM applied in distributed fixed-time observer-to-controller channel [36], the designed scheme optimizes resource utilization by discontinuous output, intermittent connectivity, and aperiodic execution.

V. Simulation Results

In this section, simulation and experiment are conducted to demonstrate the validity of the developed DFTFCC algorithm for uncertain nonlinear MASs.

A. Numerical Simulation

Note that the dynamic system (1) can represent the outer-loop subsystem of a quadrotor UAV [1]. Consider the networked MASs containing one virtual leader labeled as 0, four formation leader UAVs labeled as 1–4, and four follower UAVs labeled as 5–8. Note that the control input of the virtual leader is $u_0 = [0; -(\pi^2/20)\sin(\pi t/20); 0]$. And the communication topology among multi-UAV is given in Fig. 2. The initialization parameters of entire control system are set to $\beta_1 = \beta_2 = \lambda_1 = \lambda_2 = 0.2, \alpha_1^* = \alpha_2^* = 1.5, \gamma_1 = \gamma_3 = 3, \gamma_2 = \gamma_4 = 1.2, \xi_{i,l\eta}^1 = 0.001, \xi_{i,l\eta}^2 = 0.01. L_{i,v} = 15, \kappa_{i,2} = \kappa_{i,1} = \kappa_{i,3} = 15, m = 0.1, T_u = 0.1. k_{i,p} = 0.5, k_{i,v} = 1, p = 0.8,\ q = 1.2, \kappa = 0.5, c_{i,1} = c_{i,1}^{'} = c_{i,2} = \tilde{c}_{i,2} = 0.05$.

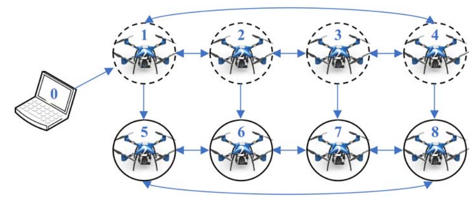

Fig. 2. Communication topology.

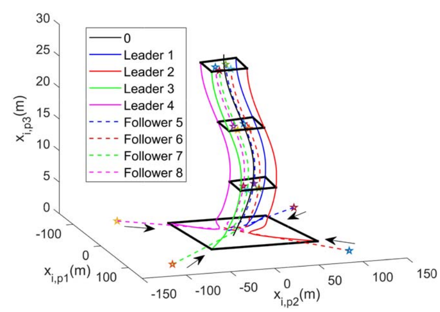

Fig. 3. Trajectories of formation leaders and followers under proposed control scheme.

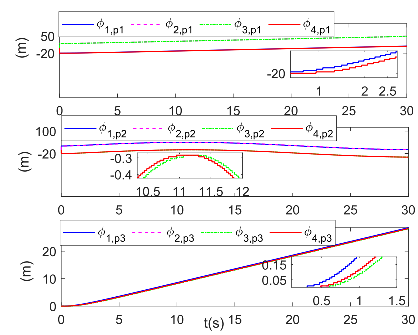

Fig. 4. Evolution of DFTETE states (5).

Simulations are performed in MATLAB/Simulink by setting a fixed sampling step as 0.001 s. Fig. 3 shows the 3-D trajectory of formation leaders and followers under the proposed control strategy, where we can see that the formation leaders realize the predefined formation pattern and the followers are driven into the geometric region spanned by the leaders with network connectivity. The time evolution of the proposed FTETE states is shown in Fig. 4, where we can observer that the transmission of estimator states among neighbors is not continuous. From Figs. 5 and 6, the formation errors and containment errors converge to a small domain around zero within 7 s regardless of different initial errors, which implies that the DFTFCC is effective despite reducing the frequency of communication and controller update. The control inputs of formation leaders and followers are shown in Figs. 7 and 8. Additionally, the effect of the proposed ETMs is quantitatively analyzed by using the collected data within 30 s. Using the information transmission of the formation leader 1 as an illustration, the number of triggering events under our developed strategy is 7050 and 5785 for

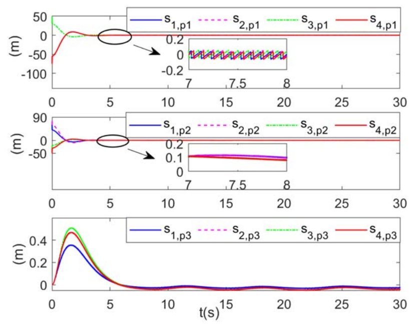

Fig. 5. Time curves of formation errors of the formation leaders.

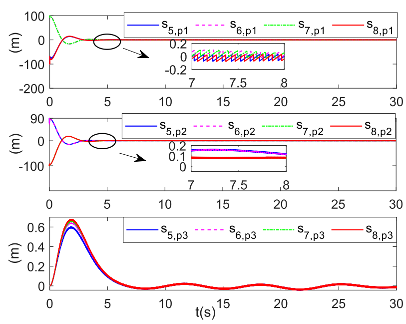

Fig. 6. Time curves of containment errors of the followers.

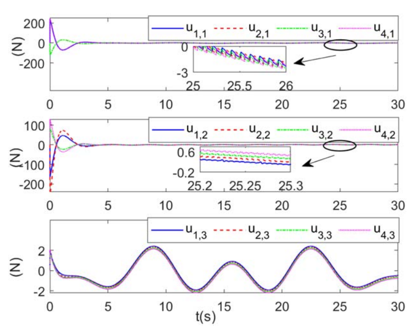

Fig. 7. Time curves of control inputs of the formation leaders.

$\hat{x}_{1,p\eta}$ and $\hat{x}_{1,v\eta}$ among neighbors communication, 31 318 and 24 143 for $x_{1,p\eta}$ and $\alpha_{1,1f\eta}$ among sensor-to-controller channels, 41 429 for $v_{1,\eta}$ among controller-to-actuator channels. The triggering rate means the ratio of the triggering number to the period sampling number within a time interval. The triggering rates for the mentioned signals are 7.8%, 6.4%, 34.8%, 26.8%, and 46.0%, respectively.

To interpret the strength of the proposed DFTFCC scheme in terms of reducing communication cost without sacrificing tracking accuracy, the comparisons are performed with a typical fixed-time event-triggered mechanism [25]. To relieve the conflict of the control framework and enable a fair comparison, the event trigger [25] embedded in the controller-to-actuator channel only is introduced in our control strategy. In Fig. 9, the practical fixed-time stabilization of formation errors is realized after 7 s. Moreover, the quantitative indexes including

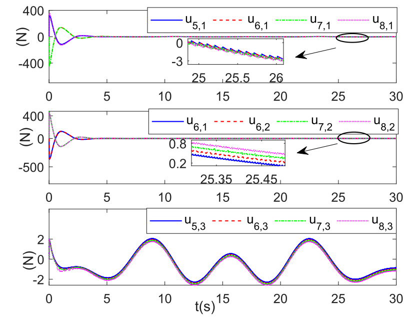

Fig. 8. Time curves of control inputs of the followers.

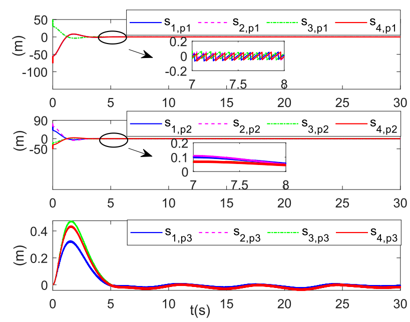

Fig. 9. Formation errors of the method in [25].

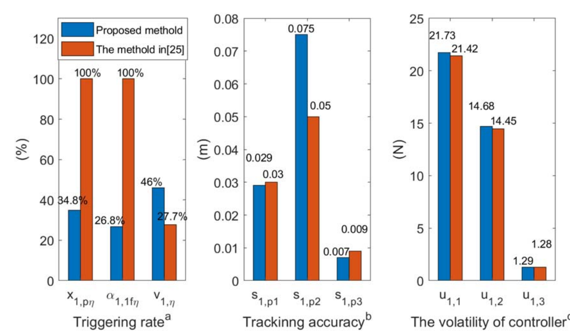

Fig. 10. Performance comparison with the method in [25]. (a) Triggering rate represents the rate of the triggering number to the period sampling number. (b) Tracking accuracy is the standard deviation of $s_{1,p\eta}(\eta = 1, 2, 3)$. (c) Volatility of controller is the standard deviation of $u_{1,\eta}(\eta = 1, 2, 3)$.

triggering rate, tracking precision and the volatility of controller are characterized in Fig. 10. By comparison, it can be found that the proposed control algorithm obtains similar control precision, while the communication resource is saved overall. In addition, the volatility of control input $\mathbf{u}_i$ in our control scheme is slightly larger than those in the scheme [25], which is the acceptable price of reducing the communication between sensor-to-controller channel.

B. Experimental Simulation

A practical experiment is conducted to further validate the efficacy of the proposed scheme. The experiment platform, as illustrated in Fig. 11(a), comprises an optical capturer system, a

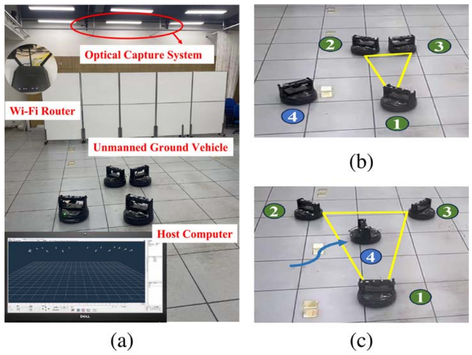

Fig. 11. (a) Experimental platform. (b) Initial positions of UGVs. (c) Results of formation-containment.

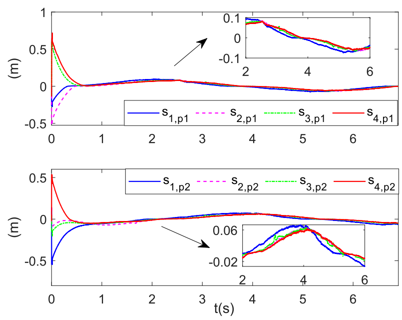

Fig. 12. Tracking errors in experimental simulation.

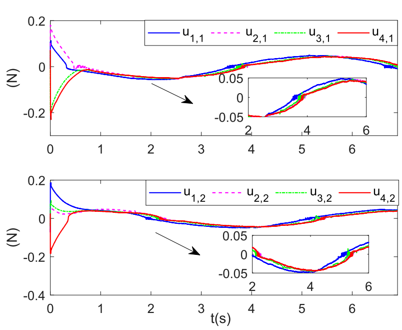

Fig. 13. Control inputs of UGVs in experimental simulation.

Wi-Fi router, a host computer, and four Qbot2 UGVs containing three formation leader UGVs labeled 1–3 and one follower UGV labeled 4. Notably, the positions of the UGVs are tracked in real-time using a motion capture system, and the resulting measurement data is transmitted to the host computer via the Wi-Fi router. With the received information and the predefined formation-containment pattern, these commands are transmitted for execution to the drive boards of the UGVs. Note that the control input of the virtual leader is $u_0 = [-\cos(t); -\sin(t)]$.

In the sequel, the experimental results of UGVs are shown in Fig. 11(b) and 11(c), where the formation leaders shape a predefined geometrical configuration and follower 4 reaches the convex hull. Meanwhile, the tracking errors $s_{\mathrm{ip}\eta}$ and control inputs of all UGVs are plotted in Figs. 12 and 13. Moreover, the experimental video of the formation-containment of UGVs is provided in https://youtu.be/ohbv4-4EyVM.

VI. Conclusion

In this article, a DFTFCC protocol for uncertain nonlinear MASs is proposed under the backstepping framework. First, the challenges of scarce communication bandwidth among neighbors are addressed by developing a DFTETE. This estimator effectively captures the desired position and velocity commands for both formation leaders and followers within a fixed time. A FTESO is introduced to estimate the unmeasured velocity and lumped disturbance simultaneously. Then, a fixed-time backstepping controller with aperiodic actuation is designed based on intermittent output signals. As a result, the proposed algorithm not only achieves the control objective of DFTFCC but also optimizes resource utilization. In the future, we will explore fixed-time time-varying formation control solutions for uncertain nonlinear MASs with fading channel [37].

Appendix A
Proof of Theorem 1

As to the limitation of space, part of the specific deviations are omitted in this section. Consider the estimator errors as $z_{i,p\eta} = \hat{x}_{i,p\eta} - x_{0,p\eta} - \delta_{i,0\eta}, \mathbf{z}_p = [z_{1,p}^T, \ldots, z_{N,p}^T]^T$ and $z_{i,v\eta} = \hat{x}_{i,v\eta} - \dot{x}_{0,p\eta}, \mathbf{z}_v = [z_{1,v}^T, \ldots, z_{N,v}^T]^T$. Then, define

$$ee_{i,v\eta}^1 = \dot{\hat{x}}_{i,v\eta} + \beta_1^{-\frac{1}{\gamma_2}} \mathrm{sig}^{\frac{1}{\gamma_2}}(\bar{y}_{i,v\eta} + \alpha_1 \mathrm{sig}^{\gamma_1}(\bar{y}_{i,v\eta})) + \lambda_1 \mathrm{sign}(\bar{y}_{i,v\eta})$$
$$\bar{y}_{i,v\eta} = \sum_{j=0}^{N} a_{ij}(\hat{x}_{i,v\eta}(t) - \hat{x}_{j,v\eta}(t)) \tag{26}$$

where $\alpha_1^* = [(2^{-(\gamma_1-\gamma_2/\gamma_2)}\alpha_1 - 2^{-(\gamma_1-\gamma_2/\gamma_2)}\alpha_1)/2]\mathrm{sign}(\hat{x}_{i,v\eta}(t) - \hat{x}_{j,v\eta}(t_{j,v\eta}^{k'})) + [(2^{-(\gamma_1-\gamma_2/\gamma_2)}\alpha_1 + 2^{-(\gamma_1-\gamma_2/\gamma_2)}\alpha_1)/2]$.

Subsequently, uncertainties induced of triggering event can be expressed as $\Delta_{i,v\eta} = e_{i,v\eta}^1 + ee_{i,v\eta}^1$. Based on the graph theory, the time differential of estimator error is rewritten as

$$\begin{aligned}\dot{\mathbf{z}}_v = &-\beta_1^{-\frac{1}{\gamma_2}} \mathrm{sig}^{\frac{1}{\gamma_2}}\left[(\mathcal{L}_{L1} \otimes \mathbf{I}_3)\mathbf{z}_v + \alpha_1 \mathrm{sig}^{\gamma_1}((\mathcal{L}_{L1} \otimes \mathbf{I}_3)\mathbf{z}_v)\right] \\ &- \lambda_1 \mathrm{sign}\left[(\mathcal{L}_{L1} \otimes \mathbf{I}_3)\mathbf{z}_v\right] - 1_N \otimes \dot{\mathbf{X}}_{0,v} + (\mathbf{e}_v^1 + \mathbf{ee}_v^1)\end{aligned}$$

where $\mathbf{e}_v^1 = [{\mathbf{e}_{1,v}^1}^T, \ldots, {\mathbf{e}_{N,v}^1}^T]^T, \mathbf{ee}_v^1 = [{\mathbf{ee}_{1,v}^1}^T, \ldots, {\mathbf{ee}_{N,v}^1}^T]^T$; $\otimes$ is the Kronecker product and $1_N = [1, \ldots, 1]^T$.

With the aid of the rule of the Kronecker product and Graph theory, it can be concluded that $\mathcal{L}_{L1} \otimes \mathbf{I}_3$ is positive definite. Then, construct the Lyapunov function $\mathbf{V}_1 = (1/2){\mathbf{z}_v}^T(\mathcal{L}_{L1} \otimes \mathbf{I}_3)\mathbf{z}_v$ and define $\rho = (\mathcal{L}_{L1} \otimes \mathbf{I}_3)\mathbf{z}_v$ with $\rho = [\rho_{11}, \rho_{12}, \ldots, \rho_{N2}, \rho_{N3}]^T$.

According to the predefined ETMs and Lemma 3, we have $e_{i,v\eta}^1 \leq \xi_{i,v\eta}^1$ and $\left|ee_{i,v\eta}^1\right| \leq \Gamma_{i,v\eta}$ with $\Gamma_{i,v\eta}$ being a positive constant. Then, we can conduct that

$$\begin{aligned}\dot{\mathbf{V}}_1 \leq &-\beta_1^{-\frac{1}{\gamma_2}}\left(\sum_{i=1}^{N}\sum_{\eta=1}^{3}\left|{\rho_{i\eta}}^2\right|^{\frac{\gamma_2+1}{2}} + \sum_{i=1}^{N}\sum_{\eta=1}^{3}\alpha_1({\rho_{i\eta}}^2)^{\frac{\gamma_2+\gamma_1}{2}}\right)^{\frac{1}{\gamma_2}} \\ &+ \left((3n)^{-\frac{1}{2}}(\Xi + \xi_{i,v\eta}^1 + \Gamma_{i,v\eta} - \lambda_2)\left(\sum_{i=1}^{N}\sum_{\eta=1}^{3}\left|{\rho_{i\eta}}^2\right|\right)^{\frac{\gamma_2}{2}}\right)^{\frac{1}{\gamma_2}} \\ \leq &-\left(\frac{\varpi_1}{\beta_1}V_1^{\frac{\gamma_2+1}{2}} + \frac{\alpha_1}{\beta_1}\varpi_2 V_1^{\frac{\gamma_2+\gamma_1}{2}} - \bar{\varpi}_1 V_1^{\frac{\gamma_2}{2}}\right)^{\frac{1}{\gamma_2}}\end{aligned} \tag{27}$$

where

$$\varpi_1 = (3N)^{\frac{1-\gamma_2}{2}}(2\lambda_{\min}(\mathcal{L}_{L1}))^{\frac{1+\gamma_2}{2}}$$
$$\varpi_2 = (3N)^{\frac{2-\gamma_1-\gamma_2}{2}}(2\lambda_{\min}(\mathcal{L}_{L1}))^{\frac{\gamma_1+\gamma_2}{2}}$$
$$\bar{\varpi}_1 = (3N)^{-\frac{1}{2}}(\Xi + \xi^1_{i,v\eta} + \Gamma_{i,v\eta} - \lambda_2)(2\lambda_{\max}(\mathcal{L}_{L1}))^{\frac{\gamma_2}{2}}.$$

Assuming that $(\varpi_1/\beta_1)\psi V_1^{(\gamma_2+1/2)} \geq \bar{\varpi}_1 V_1^{(\gamma_2/2)}$with $\psi \in (0,1)$, we have

$$\dot{\mathbf{V}}_1 \leq -\left(\frac{\varpi_1}{\beta_1}(1-\psi)V_1^{\frac{\gamma_2+1}{2}} + \frac{\alpha_1}{\beta_1}\varpi_2 V_1^{\frac{\gamma_2+\gamma_1}{2}}\right)^{\frac{1}{\gamma_2}}. \quad (28)$$

Recalling Lemma 2, the powers of (28) satisfy $(\gamma_2 + 1/2)(1/\gamma_2) < 1, (\gamma_2 + \gamma_1/2)(1/\gamma_2) > 1$. Thus, we can deduce that the developed DFTETE can achieve practical fixed-time convergence. And the setting time is given as

$$T_e \leq \frac{1}{\left(\frac{\varpi_1}{\beta_1}(1-\psi)\right)^{\frac{1}{\gamma_2}}\left(1-\frac{\gamma_2+1}{2}\frac{1}{\gamma_2}\right)} + \frac{1}{\left(\frac{\alpha_1\varpi_2}{\beta_1}\right)^{\frac{1}{\gamma_2}}\left(\frac{\gamma_2+\gamma_1}{2}\frac{1}{\gamma_2}-1\right)}.$$

Similarly, the Lyapunov function for the estimator error dynamics of position loop is designed as $\mathbf{V}_2 = (1/2)\mathbf{z}_p{}^T(\mathcal{L}_{L1} \otimes \mathbf{I}_3)\mathbf{z}_p$, and the command of $\mathbf{X}_{0,p} + \delta_{i,0}$ can be obtained locally via $\hat{\mathbf{X}}_{i,p}$ within T_e. Furthermore, we introduce the following proof process to exclude the Zeno problem:

Differentiating the measurement error yields

$$\left|de^1_{i,v\eta}/dt\right| \leq \left(\beta_1^{-\frac{1}{\gamma_2}}\left|\frac{1}{\gamma_2}\left|\bar{y}^e_{i,v\eta}\right|^{\frac{1-\gamma_2}{\gamma_2}} + \alpha_1\frac{\gamma_1}{\gamma_2}\left|\bar{y}^e_{i,v\eta}\right|^{\frac{\gamma_1-\gamma_2}{\gamma_2}}\right| + 2\lambda_1\right)\left|\dot{\hat{x}}_{i,v\eta}(t)\right|$$
$$\bar{y}^e_{i,v\eta} = \sum_{j=0}^{N} a_{ij}(\hat{x}_{i,v\eta}(t) - \hat{x}_{j,v\eta}(t) - \sum_{j=0}^{N} a_{ij}e^2_{j,v\eta}). \quad (29)$$

Moreover, recalling the above stability analysis, since the velocity of the virtual leader is bounded and the estimator error also exist a supper bound, we can easily know that there must exist a constant $\pi_{i,v1}$ such that $\left|de^1_{i,v\eta}/dt\right| \leq \pi_{i,v1}$.

Recalling preset event-triggered conditions, we have

$$\left|e^1_{i,v\eta}(t^{k+1}_{i,v\eta})\right| = \xi^1_{i,v\eta} \leq \int_{t^k_{i,v\eta}}^{t^{k+1}_{i,v\eta}} \pi_{i,v1}ds = \pi_{i,v1}(t^{k+1}_{i,v\eta} - t^k_{i,v\eta}).$$

Then,

$$t^{k+1}_{i,v\eta} - t^k_{i,v\eta} \geq \xi^1_{i,v\eta}/\pi_{i,v1}.$$

Similarly, the proof of no Zeno behavior for $e^1_{i,p\eta}$, $e^2_{i,p\eta}$, and $e^2_{i,v\eta}$ is ensured. ■

Appendix B
Proof of Theorem 2

According to controllers (17) and (21), we have

$$\begin{aligned}\dot{\mathbf{s}}_{i,p} &= \mathbf{s}_{i,v} + \Delta_i + \bar{\alpha}_{i,1} + (\alpha_{i,1} - \bar{\alpha}_{i,1}) - \dot{\phi}_{i,p}(t) + \mathbf{Z}_{i,v}\\ &= \mathbf{s}_{i,v} - k_{i,p}[\breve{\mathbf{s}}_{i,p} + \mathrm{sig}^p(\breve{\mathbf{s}}_{i,p}) + \mathrm{sig}^q(\breve{\mathbf{s}}_{i,p})]\\ &\quad + \alpha_{i,1} - \bar{\alpha}_{i,1} + \Delta_i + \mathbf{Z}_{i,v}\end{aligned} \quad (30)$$

$$\begin{aligned}\dot{\mathbf{s}}_{i,v} &= (u-v) - (\breve{\mathbf{s}}_{i,p} - \mathbf{s}_{i,p}) - k_{i,v}[\breve{\mathbf{s}}_{i,v} + \mathrm{sig}^p(\breve{\mathbf{s}}_{i,v}) + \mathrm{sig}^q(\breve{\mathbf{s}}_{i,v})]\\ &\quad + (\bar{\alpha}_{i,1} - \breve{\alpha}_{i,1f})/\kappa - \mathbf{s}_{i,p} - \dot{\alpha}_{i,1f} - \dot{\mathbf{Z}}_{i.v} + \mathbf{Z}_{i,\varphi}.\end{aligned} \quad (31)$$

Recalling the observer error dynamics (8), there exist positive constants T_{io} and $\bar{h}_i$ [34], such that $\forall t > T_{i,o}$, $\left\|\dot{\mathbf{Z}}_{i.v}\right\| + \|\mathbf{Z}_{i,v}\| + \|\mathbf{Z}_{i,\varphi}\| \leq \bar{h}_i$.

Since for any $\Gamma_1 > 0$, the set $\Omega_1 = \{\mathbf{s}^T_{i,p}\mathbf{s}_{i,p} + \mathbf{s}^T_{i,v}\mathbf{s}_{i,v} + \mathbf{Z}^T_{i,p}\mathbf{Z}_{i,p} + \mathbf{Z}^T_{i,v}\mathbf{Z}_{i,v} + \mathbf{Z}^T_{i,\varphi}\mathbf{Z}_{i,\varphi} + \Delta_i{}^T\Delta_i \leq 2\Gamma_1\}$ is a compact set. Therefore, there exists a positive constant ϖ ensuring $\|-\dot{\alpha}_1\|^2 \leq \varpi$ on Ω_1. For filter error, design the Lyapunov function as $V(\Delta_i) = (1/2)\Delta_i{}^T\Delta_i$, and its derivative satisfies

$$\dot{V}(\Delta_i) = \Delta_i\left(-\frac{\Delta_i}{\kappa} - \dot{\alpha}_1\right) \leq -\left(\frac{1}{\kappa} - \frac{1}{4}\right)\|\Delta_i\|^2 + \varpi \quad (32)$$

where $(1/\kappa) - (1/4) > 0$. Inequality (32) means that $\dot{V}(\Delta_i) < 0$ on $V(\Delta_i) = \varrho\Gamma_1, \varrho \in (0,1)$ when $(1/\kappa) > (\varpi/2\varrho\Gamma_1) + (1/4)$. Hence, $V(\Delta_i)$ is an invariant set. it can be concluded that the filter error is semiglobally bounded. Thus, we have $\left\|\Delta_i{}^2\right\| + 2\bar{h}_i^2 \leq B$ with B being a constant.

Another Lyapunov function is designed as $V_{3,i} = (1/2)\mathbf{s}^T_{i,p}\mathbf{s}_{i,p} + (1/2)\mathbf{s}^T_{i,v}\mathbf{s}_{i,v}$. Based on the Lemmas 3 and 4, we get the following inequality:

$$\begin{aligned}\mathbf{s}^T_{i,p}\dot{\mathbf{s}}_{i,p} &\leq \mathbf{s}^T_{i,p}\mathbf{s}_{i,v} - k_{i,p}\left(1 - \frac{1+2c_{i,1}}{2k_{i,p}} - c_{i,1}\right)\left\|\breve{\mathbf{s}}_{i,p}\right\|^2 + \bar{h}_i^2\\ &\quad - k_{i,p}\left(1 - c_{i,1}3^{(1-p)/2} - \frac{2+c_{i,1}}{k_{i,p}}\right)\left\|\breve{\mathbf{s}}_{i,p}\right\|^{p+1}\\ &\quad - k_{i,p}\left(3^{(1-q)/2} - c_{i,1} - \frac{2+c_{i,1}}{k_{i,p}}\right)\left\|\breve{\mathbf{s}}_{i,p}\right\|^{q+1} + \|\Delta_i\|^2.\end{aligned} \quad (33)$$

Similarly

$$\begin{aligned}\mathbf{s}^T_{i,v}\dot{\mathbf{s}}_{i,v} &\leq -\mathbf{s}^T_{i,v}\mathbf{s}_{i,p} - k_{i,v}(1 - c'_{i,1} - 3(c'_{i,1}+1)^2/4k_{i,v}\\ &\quad - (c'_{i,1}+1)(c'_{i,1} + c'_{i,2}\kappa)/\kappa k_{i,v})\left\|\breve{\mathbf{s}}_{i,v}\right\|^2\\ &\quad - k_{i,v}\left(1 - c'_{i,1}3^{(1-p)/2} - \frac{(c'_{i,1}+1)((2+c_{i,1})r}{\kappa k_{i,v}(p+1)}\right)\left\|\breve{\mathbf{s}}_{i,v}\right\|^{p+1}\\ &\quad - k_{i,p}\left((3)^{(1-q)/2} - c'_{i,1} - \frac{(c'_{i,1}+1)((2+c_{i,1})r}{\kappa k_{i,v}(q+1)}\right)\left\|\breve{\mathbf{s}}_{i,v}\right\|^{q+1}\\ &\quad + (c_{i,1}{}^2 + \kappa^2 c_{i,1}{}^2/\kappa^2)\left\|\breve{\mathbf{s}}_{i,p}\right\|^2 + \frac{(c'_{i,1}+1)((2+c_{i,1})p}{\kappa(p+1)}r^{\frac{-1}{p}}\\ &\quad \times \left\|\breve{\mathbf{s}}_{i,p}\right\|^{p+1} + \frac{(c'_{i,1}+1)((2+c_{i,1})q}{\kappa(q+1)}r^{\frac{-1}{q}}\left\|\breve{\mathbf{s}}_{i,p}\right\|^{q+1} + \bar{h}_i^2.\end{aligned} \quad (34)$$

Following (33) and (34), we have the derivative of $V_{3,i}$ with respect to time

$$\begin{aligned}\dot{V}_{3,i} &\leq -\rho_{i,1}\left(\left\|\breve{\mathbf{s}}_{i,p}\right\|^2 + \left\|\breve{\mathbf{s}}_{i,v}\right\|^2\right)\\ &\quad - \rho_{i,2}\left(\left\|\breve{\mathbf{s}}_{i,p}\right\|^{p+1} + \left\|\breve{\mathbf{s}}_{i,v}\right\|^{p+1}\right)\\ &\quad - \rho_{i,3}\left(\left\|\breve{\mathbf{s}}_{i,p}\right\|^{p+1} + \left\|\breve{\mathbf{s}}_{i,v}\right\|^{q+1}\right) + B\end{aligned} \quad (35)$$

where $\bar{\rho}_{i,1}, \bar{\rho}_{i,2}$, and $\bar{\rho}_{i,3}$ are the positive constants associated with the controller parameters.

Given $\bar{\varepsilon}_i(t) = [\mathbf{s}_{i,p}{}^T, \mathbf{s}_{i,v}{}^T]^T$, we have $\bar{\phi}_i(t) = \mathbf{s}_{i,p}{}^T + \mathbf{s}_{i,v}{}^T = (1,1) \otimes I_3 \bar{\varepsilon}_i(t)$. Before formation control objective is realized, we have

$$\bar{\phi}_i(t)^T \bar{\phi}_i(t) \geq \omega(\mathbf{s}_{i,p}^T \mathbf{s}_{i,p} + \mathbf{s}_{i,v}^T \mathbf{s}_{i,v}) \tag{36}$$

where $\omega = \min_{(\bar{\varepsilon}(t)/\|\bar{\varepsilon}(t)\|)} (\bar{\varepsilon}(t)/\|\bar{\varepsilon}(t)\|)^T \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \otimes I_3 (\bar{\varepsilon}(t)/\|\bar{\varepsilon}(t)\|) > 0, \bar{\varepsilon}(t) = [\bar{\varepsilon}_1^T, \bar{\varepsilon}_2^T]^T$.

According to Jensen Inequality, we have

$$\begin{aligned} -\dot{V}_{3,i} &\geq 2\bar{\rho}_{i,2}\left(\frac{\|\mathbf{s}_{i,p}+\mathbf{s}_{i,v}\|}{2}\right)^{p+1} + 2\bar{\rho}_{i,3}\left(\frac{\|\mathbf{s}_{i,p}+\mathbf{s}_{i,v}\|}{2}\right)^{q+1} - B \\ &= 2^{\frac{1-p}{2}}\omega\bar{\rho}_{i,2} V_{3,i}{}^{(p+1)/2} + 2^{\frac{1-q}{2}}\omega\bar{\rho}_{i,3} V_{3,i}{}^{(q+1)/2} - B. \end{aligned} \tag{37}$$

As a result, we can obtain the reaching time T_c

$$T_c = \frac{2}{\sum_{i=1}^N 2^{\frac{1-p}{2}}\omega\bar{\rho}_{i,2}(1-p)} + \frac{2}{\sum_{i=1}^N 2^{\frac{1-q}{2}}\omega\bar{\rho}_{i,3}(q-1)}. \tag{38}$$

Then, the error signal $\mathbf{s}_{i,p}, \mathbf{s}_{i,v} (i \in \mathcal{V}_l)$ converge to compact sets after $t = T_e + T_c$. Recalling the Lemma 2, it is deduced that the practical fixed-time stabilization for formation leaders is established using the intermittent output, such that the formation leaders form a predefined formation pattern and move dynamically along with virtual leader's directive.

Subsequently, the Zeno phenomenon will be excluded with the aid of the following analysis. From (37), we conclude that all states are upper-bounded as $|x_{i,v\eta}| \leq \bar{\upsilon}$, where $\bar{\upsilon}$ is a positive constant.

The output triggering condition satisfies

$$d\left|\breve{x}_{i,p\eta} - x_{i,p\eta}\right|/dt = \text{sign}(\breve{x}_{i,p\eta} - x_{i,p\eta})\dot{x}_{i,p\eta}(t) \leq |x_{i,v\eta}(t)|. \tag{39}$$

Since $\breve{x}_{i,p\eta}(t) - x_{i,p\eta}(t^\sigma_{i,p\eta}) = 0$, following (39), we have

$$\left|\breve{x}_{i,p\eta}(t) - x_{i,p\eta}(t)\right| \leq \int_{t^\sigma_{i,p\eta}}^t |x_{i,v\eta}(t)| ds \leq \bar{\upsilon}(t - t^\sigma_{i,p\eta}). \tag{40}$$

According to the triggering limit (16) and (40), we get

$$\left|\breve{x}_{i,p\eta}(t) - x_{i,p\eta}(t^{\sigma+1}_{i,p\eta})\right| = c_{i,1}\left|\breve{\mathbf{s}}_{i,p\eta}(t^{\sigma+1}_{i,p\eta})\right| \leq \int_{t^\sigma_{i,p\eta}}^{t^{\sigma+1}_{i,p\eta}} |x_{i,v\eta}(t)| \mathrm{d}s \tag{41}$$

which yields

$$t^{\sigma+1}_{i,p\eta} - t^\sigma_{i,p\eta} > \frac{c_{i,1}\left|\breve{\mathbf{s}}_{i,p\eta}(t^{\sigma+1}_{i,p\eta})\right|}{\bar{\upsilon}} > 0. \tag{42}$$

The practical fixed-time stability is established through Theorem 2, implying that all errors will converge to a compact set around zero. According to the definition of zero behavior, the discontinuous errors are set as a threshold and the zero behavior can be excluded even upon achieving the desired formation. Additionally, the controller update relies on the discontinuous error signal without Zeno sampling, and the triggering threshold governing the control signal relay exists positive boundedness. ■

References

[1] X. Dong, Y. Hua, Y. Zhou, Z. Ren, and Y. Zhong, "Theory and experiment on formation-containment control of multiple multirotor unmanned aerial vehicle systems," *IEEE Trans. Autom. Sci. Eng.*, vol. 16, no. 1, pp. 229–240, Jan. 2019.

[2] Y. Yang, X. Si, D. Yue, and J. Tan, "Time-varying formation tracking of uncertain nonaffine nonlinear multiagent systems with communication delays," *IEEE Trans. Ind. Electron.*, vol. 68, no. 3, pp. 2501–2509, Mar. 2021.

[3] H. Wang and J. Shan, "Fully distributed event-triggered formation control for multiple quadrotors," *IEEE Trans. Ind. Electron.*, vol. 70, no. 12, pp. 12566–12575, Dec. 2023.

[4] M. Lv, B. De Schutter, and S. Baldi, "Nonrecursive control for formation-containment of HFV swarms with dynamic event-triggered communication," *IEEE Trans. Ind. Informat.*, vol. 19, no. 3, pp. 3188–3197, Mar. 2023.

[5] W. Xiao, H. Ma, L. Zhou, and H. Li, "Adaptive fuzzy fixed-time formation-containment control for Euler-Lagrange systems," *IEEE Trans. Fuzzy Syst.*, vol. 31, no. 10, pp. 3700–3709, Oct. 2023.

[6] J. Duan, G. Duan, S. Cheng, S. Cao, and G. Wang, "Fixed-time time-varying output formation-containment control of heterogeneous general multi-agent systems," *ISA Trans.*, vol 137, pp. 210–221, Jun. 2023.

[7] Y. Li, Y. Zhao, W. Liu, and J. Hu, "Adaptive fuzzy predefined-time control for third-order heterogeneous vehicular platoon systems with dead zone," *IEEE Trans. Ind. Informat.*, vol. 19, no. 9, pp. 9525–9534, Sep. 2023.

[8] S. Dai, Z. Wu, P. Zhang, M. Tan, and J. Yu, "Distributed formation control for a multirobotic fish system with model-based event-triggered communication mechanism," *IEEE Trans. Ind. Electron.*, vol. 70, no. 11, pp. 11433–11442, Nov. 2023.

[9] Z. Han, W. Wang, J. Huang, and Z. Wang, "Distributed adaptive formation tracking control of mobile robots with event-triggered communication and denial-of-service attacks," *IEEE Trans. Ind. Electron.*, vol. 70, no. 4, pp. 4077–4087, Apr. 2023.

[10] G. Cui, H. Xu, J. Yu, and H.-K. Lam, "Event-triggered distributed fixed-time adaptive attitude control with prescribed performance for multiple QUAVs," *IEEE Trans. Autom. Sci. Eng.*, early access, doi: 10.1109/TASE.2023.3297235.

[11] W. Jiang, G. Wen, Z. Peng, T. Huang, and A. Rahmani, "Fully distributed formation-containment control of heterogeneous linear multiagent systems," *IEEE Trans. Autom. Control,* vol. 64, no. 9, pp. 3889–3896, Sep. 2019.

[12] K. J. Astrom and B. M. Bernhardsson, "Comparison of Riemann and Lebesgue sampling for first order stochastic systems," in *Proc. 41st IEEE Conf. Decis. Control,* Las Vegas, NV, USA, 2002, pp. 2011–2016.

[13] Y. Qian, L. Liu, and G. Feng, "Cooperative output regulation of linear multiagent systems: An event-triggered adaptive distributed observer approach," *IEEE Trans. Autom. Control,* vol. 66, no. 2, pp. 833–840, Feb. 2021.

[14] C. Deng, C. Wen, J. Huang, X.-M. Zhang, and Y. Zou, "Distributed observer-based cooperative control approach for uncertain nonlinear MASs under event-triggered communication," *IEEE Trans. Autom. Control,* vol. 67, no. 5, pp. 2669–2676, May 2022.

[15] A. Polyakov, "Nonlinear feedback design for fixed-time stabilization of linear control systems," *IEEE Trans. Autom. Control,* vol. 57, no. 8, pp. 2106–2110, Aug. 2012.

[16] C. Zhu, Y. Jiang, and C. Yang, "Fixed-time neural control of robot manipulator with global stability and guaranteed transient performance," *IEEE Trans. Ind. Electron.*, vol. 70, no. 1, pp. 803–812, Jan. 2023.

[17] B. Tian, X. Shao, W. Yang, and W. Zhang, "Fixed time output feedback containment for uncertain nonlinear multiagent systems with switching communication topologies," *ISA Trans.*, vol. 111, pp 82–95, Apr. 2021.

[18] Y. Dong and Z. Chen, "Fixed-time synchronization of networked uncertain Euler-Lagrange systems," *Automatica,* vol. 146, Dec. 2022 Art. no. 110571.

[19] W. Li, H. Zhang, Z. Gao, Y. Wang, and J. Sun, "Fully distributed event/self-triggered bipartite output formation-containment tracking

control for heterogeneous multiagent systems," *IEEE Trans. Neural Netw. Learn. Syst.*, early access, doi: 10.1109/TNNLS.2022.3146814.

[20] A. Girard, "Dynamic triggering mechanisms for event-triggered control," *IEEE Trans. Autom. Control,* vol. 60, no. 7, pp. 1992–1997, Jul. 2015.

[21] S.-L. Du, T. Liu, and D. W. C. Ho, "Dynamic event-triggered control for leader-following consensus of multiagent systems," *IEEE Trans. Syst., Man, Cybern., Syst.,* vol. 50, no. 9, pp. 3243–3251, Sep. 2020.

[22] Z. Zhang, C. Wen, L. Xing, and Y. Song, "Adaptive event-triggered control of uncertain nonlinear systems using intermittent output only," *IEEE Trans. Autom. Control,* vol. 67, no. 8, pp. 4218–4225, Aug. 2022.

[23] Y. Xie, Q. Ma, J. Gu, and G. Zhou, "Event-triggered fixed-time practical tracking control for flexible-joint robot," *IEEE Trans. Fuzzy Syst.,* vol. 31, no. 1, pp. 67–76, Jan. 2023.

[24] X. Chen, T. Jia, Z. Wang, X. Xie, and J. Qiu, "Practical fixed-time bipartite synchronization of uncertain coupled neural networks subject to deception attacks via dual-channel event-triggered control," *IEEE Trans. Cybern.*, early access, doi: 10.1109/TCYB.2023.3338165.

[25] J. Liu, Y. Zhang, Y. Yu, and C. Sun, "Fixed-time leader-follower consensus of networked nonlinear systems via event/self-triggered control," *IEEE Trans. Neural Netw. Learn. Syst.,* vol. 31, no. 11, pp. 5029–5037, Nov. 2020.

[26] J. Ni, P. Shi, Y. Zhao, Q. Pan, and S. Wang, "Fixed-time event-triggered output consensus tracking of high-order multiagent systems under directed interaction graphs," *IEEE Trans. Cybern.,* vol. 52, no. 7, pp. 6391–6405, Jul. 2022.

[27] D. Swaroop, J. K. Hedrick, P. P. Yip, and J. C. Gerdes, "Dynamic surface control for a class of nonlinear systems," *IEEE Trans. Autom. Control*, vol. 45, no. 10, pp. 1893–1899, Oct. 2000.

[28] J. Huang, W. Wang, C. Wen, and G. Li, "Adaptive event-triggered control of nonlinear systems with controller and parameter estimator triggering," *IEEE Trans. Autom. Control,* vol. 65, no. 1, pp. 318–324, Jan. 2020.

[29] L. Wang, C. L. P. Chen, and H. Li, "Event-triggered adaptive control of saturated nonlinear systems with time-varying partial state constraints," *IEEE Trans. Cybern.,* vol. 50, no. 4, pp. 1485–1497, Apr. 2020.

[30] J. Long, W. Wang, C. Wen, J. Huang, and J. Lü, "Output feedback based adaptive consensus tracking for uncertain heterogeneous multi-agent systems with event-triggered communication," *Automatica,* vol. 136, Feb. 2022. Art. no. 110049.

[31] L. Sun, X. Huang, and Y. Song, "Distributed event-triggered control of networked strict-feedback systems via intermittent state feedback," *IEEE Trans. Autom. Control*, early access, doi: 10.1109/TAC.2022.3217914.

[32] B. Tian, L. Liu, H. Lu, Z. Zuo, Q. Zong, and Y. Zhang, "Multivariable finite time attitude control for quadrotor UAV: Theory and experimentation," *IEEE Trans. Ind. Electron.,* vol. 65, no. 3, pp. 2567–2577, Mar. 2018.

[33] A.-M. Zou and Z. Fan, "Fixed-time attitude tracking control for rigid spacecraft without angular velocity measurements," *IEEE Trans. Ind. Electron.,* vol. 67, no. 8, pp. 6795–6805, Aug. 2020.

[34] M. T. Angulo, J. A. Moreno, and L. Fridman, "Robust exact uniformly convergent arbitrary order differentiator," *Automatica,* vol. 49, no. 8, pp. 2489–2495, Aug. 2013.

[35] A. Levant, "Higher-order sliding modes, differentiation and output-feedback control," *Int. J. Control,* vol. 76, nos. 9–10, pp. 924–941. Jan. 2003.

[36] D. Jiang, G. Wen, Z. Peng, J.-L. Wang, and T. Huang, "Fully distributed pull-based event-triggered bipartite fixed-time output control of heterogeneous systems with an active leader," *IEEE Trans. Cybern.,* vol. 53, no. 5, pp. 3089–3100, May 2023.

[37] H. Yang, H. Zhang, Z. Wang, and H. Yan, "Reliable leader-following consensus of discrete-time semi-Markovian jump multi-agent systems," *IEEE Trans. Netw. Sci. Eng.*, vol. 10, no. 6, pp. 3505–3518, Nov. 2023.

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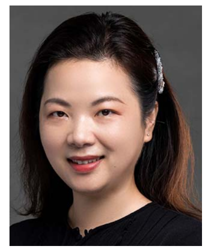

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